

# Student Attendance Management System

## Comprehensive User Manual & System Operations Guide

|                  |                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------|
| Target Audience: | School Administrators, Subject Teachers, and Academic Coordinators                                |
| Coverage Area:   | Classes XI and XII (Science, Commerce, and Arts Streams)                                          |
| Academic Year:   | 2026-27                                                                                           |
| System Edition:  | Version 1.0 (Digital Web Portal)                                                                  |
| Release Date:    | October 2026                                                                                      |
| Repository URL:  | <a href="https://github.com/ngwaradkar/attendance-">https://github.com/ngwaradkar/attendance-</a> |

**Executive Overview:** The Vidya Prabodhini Digital Attendance Portal replaces physical paper registers with a modern, paperless, real-time classroom attendance tracking system. Designed specifically for Classes XI and XII, the platform provides rapid 30-second lecture roll calls, optimistic concurrency locking for attendance corrections, comprehensive audit logging, automated defaulter identification, and formatted Excel reporting.

### Initial Demo Access Credentials

| Role               | Username | Password   | Assigned Scope                                                       |
|--------------------|----------|------------|----------------------------------------------------------------------|
| Administrator      | admin    | admin123   | Full institution access: student/teacher masters, reports, settings  |
| Teacher (Math)     | teacher1 | teacher123 | Mathematics -- Classes XI-A, XI-B, XII-A, XII-B (Science & Commerce) |
| Teacher (English)  | teacher2 | teacher123 | English -- All divisions across Science and Commerce                 |
| Teacher (Science)  | teacher3 | teacher123 | Physics & Chemistry -- Classes XI-A & XII-A                          |
| Teacher (Commerce) | teacher4 | teacher123 | Accountancy & Economics -- Classes XI-B & XII-B                      |
| Teacher (IT/CS)    | teacher5 | teacher123 | Computer Science & Information Technology                            |

## 1. System Architecture & Role-Based Access Control

The platform enforces strict role-based access control (RBAC) separating administrative governance from classroom teaching workflows:

- **Administrator Role:** Possesses institution-wide authority. Can configure academic years, classes, streams, subjects, holidays, and system thresholds; manage student and teacher masters; review school-wide daily monitors and defaulter lists; import students via Excel; inspect audit logs; and manage demo resets.
- **Teacher Role:** Scoped strictly to permitted assignments. Can view scheduled daily lectures; take classroom attendance for assigned subjects; review and correct their own recorded lectures; search student profiles; and export subject-wise attendance reports.

### Workspace Navigation Directory:

| Page Name                | Role  | Primary Purpose & Workflow                                                                     |
|--------------------------|-------|------------------------------------------------------------------------------------------------|
| Dashboard / My Dashboard | Both  | Daily attendance KPIs, lecture trends, subject breakdown, and pending timetable slots.         |
| Take Attendance          | Both  | Core classroom roll call interface. Filter by lecture, toggle absences, and save.              |
| Today's Attendance       | Admin | Live campus-wide monitoring showing enrolled vs attended counts and copyable absentee list.    |
| Attendance History       | Both  | Historical lecture registers with one-click correction, cancellation, and revision logs.       |
| Student Search           | Both  | Individual student profile search by name, roll number, or GR number with monthly trend chart. |
| Attendance Reports       | Both  | Class attendance summaries, subject matrices, and monthly P/A registers.                       |
| Defaulter Report         | Admin | Identifies students falling below configurable threshold with color-coded alerts.              |
| Student Master           | Admin | Create, update, activate/deactivate students and export current rosters.                       |
| Teacher Master           | Admin | Register teachers and configure weekly subject/period timetable slots.                         |
| Academic Master          | Admin | Manage classes, divisions, streams, subjects, academic years, and calendar holidays.           |
| Excel Import             | Admin | Batch student enrollment with automated schema validation and duplicate detection.             |
| Audit Log                | Admin | Immutable chronological security and operations audit log for all system events.               |
| Settings                 | Admin | Portal branding, threshold values, periods per day, password resets, and demo controls.        |

## 2. Teacher User Guide -- Classroom Workflows

Taking classroom attendance is optimized for mobile, tablet, and desktop touch screens with a **Default-Present** approach:

- **Step 1 -- Sign In:** Navigate to the portal and enter your teacher credentials (e.g. teacher1 / teacher123). Click **Login**.
- **Step 2 -- Review Today's Schedule:** The **My Dashboard** page displays all assigned lecture periods for today with indicators for [Pending] or [Completed].
- **Step 3 -- Open Lecture:** Click the **Open lecture** shortcut button on your scheduled period, or navigate directly to **Take Attendance**.
- **Step 4 -- Select Filters:** Choose Academic Year -> Class (e.g. XI) -> Division (e.g. XI-A) -> Stream (Science) -> Subject (Mathematics). Select the period (1-8) and ensure the date is today.
- **Step 5 -- Rapid Roll Call:** All active students automatically start toggled **Present**. Simply tap the toggle to switch off **only the absent students**. Quick-action buttons (*Mark All Present, Mark All Absent, Reset*) are provided above the roster.
- **Step 6 -- Verify Summary Cards:** Review the Present count, Absent count, and lecture percentage displayed in the top KPI cards.
- **Step 7 -- Save Attendance:** Click **Save Attendance**. A green success banner will appear listing the saved session details and the exact names of absent students.

### Making Corrections to Saved Attendance:

If a student arrives late or an absence was marked in error:

1. Open **Attendance History** (or re-select the lecture in Take Attendance).
2. Select the specific lecture session from the dropdown or click the row in the table.
3. Click **Edit Attendance** (or switch from 'View Existing' to 'Edit Attendance').
4. Adjust the student toggle from Absent to Present.
5. Enter a required **Reason for correction** (e.g. '*Late arrival verified with guardian*').
6. Click **Save Attendance**. The system logs the change in the audit trail with timestamps and user details.

### Student Profile Search & History:

- Use **Student Search** to look up any student by name, roll number, or GR number. The page shows complete attendance percentage, attended vs absent lectures, a monthly attendance trend chart, and a subject-by-subject breakdown.
- Lectures can be cancelled or restored from **Attendance History** if a timetable period was cancelled due to school assembly or weather. Cancelled lectures are preserved in history but automatically excluded from attendance percentages.

### 3. Administrator Operations Guide

Administrators possess full authority over student records, academic structures, daily operations, and system compliance:

#### Today's Attendance Monitor (Campus Overview):

The **Today's Attendance Monitor** is designed for morning roll-call oversight. It displays:

- Enrolled students vs recorded student-lecture records for each class and division.
- Real-time attendance health status: [Good] ( $\geq 85\%$ ), [Attention] ( $\geq 75\%$ ), [Low] ( $< 75\%$ ), or [No Record].
- A dedicated **Today's Absent Students** list with student names, roll numbers, and parent contact details.
- A **Copyable summary** text area formatted for rapid copy-pasting into administrative logs or parent communications.

#### Managing Master Records:

- **Student Master:** Add new students, edit contact numbers, deactivate departing students, or download full student rosters. *Note:* Historical attendance records remain permanently preserved even when a student is deactivated.
- **Teacher Master:** Create teacher user accounts and map teachers to divisions, subjects, weekdays, and periods in the weekly timetable.
- **Academic Master:** Enable or disable classes, divisions, streams, and subjects. Configure institutional calendar holidays (attendance entry is blocked on holidays, and holidays are automatically excluded from attendance percentages).

#### Defaulter Reports & Early Intervention:

The **Defaulter Report** helps coordinators spot at-risk students before final term examinations:

- Filter by academic year, class, division, stream, and monthly date range.
- Adjust the threshold dynamically (defaults to the configured 75% threshold).
- Displays critical tags: [Below Critical Threshold] ( $< 60\%$ ) or [Below Warning Threshold] ( $< 75\%$ ).
- One-click export to formatted Excel with conditional red highlight formatting.

## 4. Batch Excel Import & System Configuration

### Batch Excel Import Workflow:

To enroll entire classes at the start of the academic term without manual entry:

1. Open **Excel Import** and click **Download Student Import Template**.
2. Fill in the required 8 columns: Roll No, GR No, Student Name, Class, Division, Stream, Parent Name, Mobile.
3. Upload the workbook (.xlsx). The system performs deep validation: checking for duplicate GR numbers (case-insensitive), duplicate roll numbers within division/stream, invalid mobile characters, and unmapped classes.
4. Review the **Validation Preview** table showing Valid, Duplicate, or Failed status with exact row errors.
5. Confirm the explicit checkbox and click **Import Valid Rows**. Invalid and duplicate rows are safely skipped while valid students are created atomically.

### System Settings & Security Configuration:

- **Threshold Configuration:** Configure Good (e.g. 85%), Warning (e.g. 75%), and Critical (e.g. 60%) thresholds. The system validates that Critical < Warning <= Good <= 100%.
- **Periods per Day:** Set daily academic lecture periods (1 to 12). Default is 8 periods.
- **Institutional Identity:** Set institution name, subtitle, and custom logo image path (PNG, JPG, WebP supported).
- **Administrator Password Update:** Secure password change requiring current password verification and a minimum of 8 characters.
- **Password Security:** Uses PBKDF2-HMAC-SHA256 with 260,000 rounds and random 16-byte salt.
- **Session Security:** Uses auth epoch tracking; modifying critical settings or resetting demo data immediately invalidates all active sessions.

### Audit Trail & Compliance:

Every significant system event is recorded immutably in **Audit Log**:

- Logins and logouts with username and timestamp.
- Attendance creation, edits, and cancellations with reason and delta count.
- Student and teacher creation, modification, and deactivation.
- Searchable and exportable to formatted Excel with date filters.

## 5. Calculation Rules & Troubleshooting Guide

### Mathematical Calculation Specifications:

| Metric / Rule      | Formula & System Logic                                    | Operational Impact                                                                             |
|--------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Attendance %       | $(\text{Present} / \text{Total Recorded Lectures}) * 100$ | Percentages reflect only held lectures. Unrecorded periods are never counted as absences.      |
| Missing Records    | Excluded from denominator                                 | If a teacher is on leave or a lecture is unrecorded, students are not unfairly penalized.      |
| Holidays           | Zero records allowed; excluded from calculation           | Dates defined in Academic Master holidays cannot have attendance and do not dilute attendance. |
| Cancelled Lectures | Flagged cancelled=1; excluded from metrics                | Cancelled lectures remain visible in History for transparency but do not affect percentages.   |
| Optimistic Locking | Session timestamp revision check                          | If Teacher A and Teacher B edit the same lecture simultaneously, the stale edit is rejected.   |
| Formula Safety     | Prepends leading single quote to = + - @                  | Prevents malicious spreadsheet formula execution in Excel viewer software.                     |

### Frequently Asked Questions (FAQs):

**Q1: A teacher cannot see an assigned class in Take Attendance.**

*Resolution:* Ensure the assignment is active in **Teacher Master**. Verify that the Class, Division, Stream, Subject, and Academic Year match exactly and are marked 'Enabled' in Academic Master.

**Q2: The system displays 'This date is a holiday. Attendance is disabled.'**

*Resolution:* Attendance cannot be taken on dates listed in the institution holiday calendar. If a working day was marked as a holiday by mistake, an administrator can remove the date in **Academic Master -> Holidays**.

**Q3: An edit was rejected with 'Another user changed this lecture. Reload before editing.'**

*Resolution:* This is an optimistic concurrency safeguard. Another user modified the lecture moments prior. Simply reload the lecture to review the latest revision before making edits.

**Q4: How do we promote students from Class XI to Class XII next year?**

*Resolution:* Create the new enrollment record under Class XII using a distinct GR suffix (e.g. GR-001-XII). This preserves the student's historical Class XI attendance registers without database collision.

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**Vidya Prabodhini High School / Higher Secondary Section** | Academic Administration & IT Coordination

Official Repository: <https://github.com/ngwaradkar/attendance->